

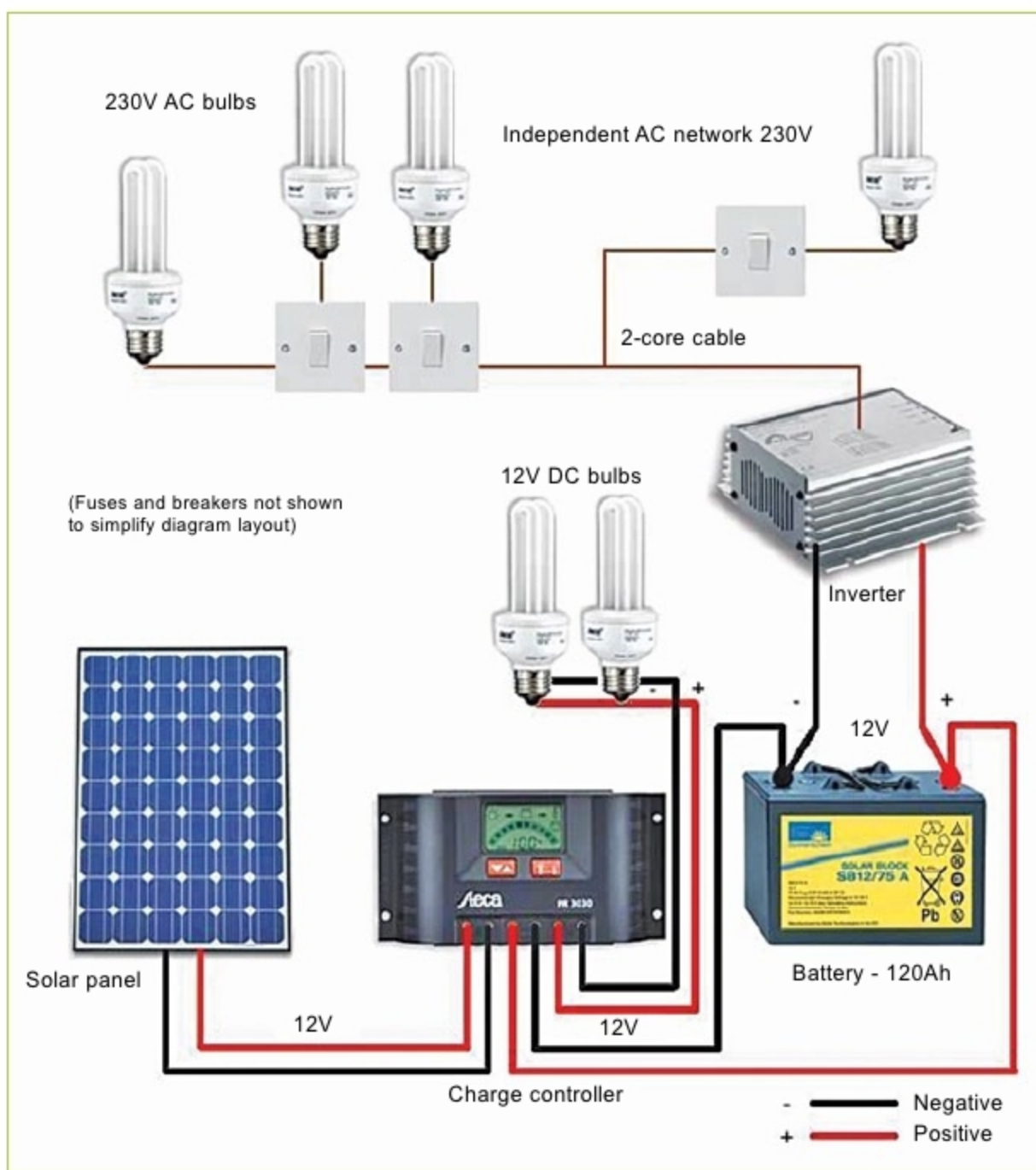


Fig. 2: Illustration showing how to set up a solar-based power backup system (Credit: jbsolis.com)

“A similar-powered solar generator running under same conditions dramatically reduces this operational cost. The only running cost associated here is for batteries, which need to be changed earliest after six years of usage.

“Solar generators run mostly by capturing solar energy. Charging at night, if required, is done through grid electricity. Even then, lithium-ion batteries get charged quickly—within three hours. So, electricity costs reduce dramatically.

“On the other hand, diesel is not required, except in emergency conditions. This is another major reduction in expense. Overall, an estimated ROI period for solar generators is about four years, depending on the extent of use, scale of application and area of implementation.”

Solar-powered generators are

future-proof, as well. Patel makes a valid point. He says, “It must be considered that prices of diesel and electricity will keep increasing with every passing year. Today, diesel costs about ₹ 70 per litre. It is only a matter of time when this becomes ₹ 100 per litre. These costs keep adding to the operational expenses. Considering that, there is a massive payback awaiting with solar.”

Market strength

While there are a good number of international vendors for solar generators, domestic production is negligibly low. However, global players are in talks to create footprints in the country. For instance, Maru Systems is ready to set up its production unit in Gujarat, within a span of one year.

Indian players like Gautam Solar and Su-Kam offer various solar-

powered power backup solutions, in addition to individual generator set components like charge controllers, batteries and panels.

Goal Zero is a popular global solar solution provider that delivers solar-powered generators for domestic and commercial uses.

India has made tremendous progress in its solar sector over the last few years. Yet, the industry is nascent and has much to deliver. Patel says, “Costs are still high, and there is limited financial assistance available from the government or banks. More assistance from the government and rising awareness will turn the tides. As domestic manufacturing increases, prices will come down and adoption rates will be higher.”

Buying tips

Investing in solar generator sets is profitable only if the area of installation frequently lacks electricity, has long-standing blackouts or is being used for critical services that must maintain zero downtime. Higher utilisation will derive quicker ROI.

Ensure that the solar generator is scalable and customisable. This allows modifications in the future, instead of having to purchase new products. Additionally, ensure that the battery is replacable. Verify the capacities of solar panels and batteries as required. Make sure that output power of the generator (maximum power capacity that it can provide for backup) is high, for multiple appliance support.

Monocrystalline solar panels are the most efficient in terms of energy generation. These are also the most space-effective, but are quite expensive. If your budget permits, go for this category of panels. Otherwise, poly-crystallines are also acceptable. These are cost-effective with moderate performance efficiency.

Check for warranty and service availability. Batteries should ideally come with three years warranty, while solar panels with 25 years warranty. **EFY**